

in IPAbet keystrokes

CONSONANTS (PULMONIC)

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b p̥ b̥		t d t̪ d̪			ʈ ɖ ʈʀ ɖʀ	c ɟ c̠ ɟ̠	k ɡ k̠ ɡ̠	q ɢ q̠ ɢ̠		ʔ ʔ̠
Nasal	m m̥	ɱ ɱ̥	n n̥			ɳ ɳ̥	ɲ ɲ̥	ŋ ŋ̥	ɴ ɴ̥		
Trill	ʙ ʙ̥		r r̥						ʀ ʀ̥		
Tap or Flap		ⱱ ⱱ̥	ɾ ɾ̥			ɽ ɽ̥					
Fricative	ɸ β ɸ̥ β̥	f v f̥ v̥	θ ð θ̪ ð̪	s z s̪ z̪	ʃ ʒ ʃ̪ ʒ̪	ɬ ɮ ɬ̪ ɮ̪	ç ʝ ç̠ ʝ̠	x ɣ x̠ ɣ̠	χ ʁ χ̠ ʁ̠	ħ ʕ ħ̠ ʕ̠	h ɦ h̠ ɦ̠
Lateral fricative			ɬ̪ ɮ̪ ɬ̪̥ ɮ̪̥								
Approximant		ʋ ʋ̥	ɹ ɹ̥			ɻ ɻ̥	j j̥	ɰ ɰ̥			
Lateral approximant			l l̥			ɭ ɭ̥	ʎ ʎ̥	ʟ ʟ̥			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

CONSONANTS (NON-PULMONIC)

Clicks

- ⊙ **pC** Bilabial
- | **tC** Dental
- ! **qC** (Post)alveolar
- ‡ **cC** Palatoalveolar
- || **lC** Alveolar lateral

Voiced implosives

b **bP** Bilabial
d **dP** Dental/alveolar
ʃ **jP** Palatal
g **gP** Velar
ŋ **qP** Uvular

Ejectives

p'  Bilabial
t'  Dental/alveolar
k'  Velar
s'  Alveolar fricative

OTHER SYMBOLS

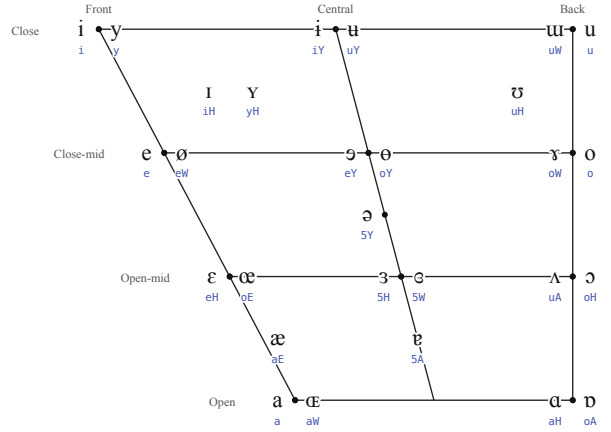
ʌ		Voiceless labial-velar fricative
W		Voiced labial-velar approximant
ɰ		Voiced labial-palatal approximant
ħ		Voiceless epiglottal fricative
ʕ		Voiced epiglottal fricative
ʡ		Epiglottal plosive

ç	[sJ]	Voiceless alveolo-palatal fricative
ž	[zJ]	Voiced alveolo-palatal fricative
ɺ	[4L]	Voiced alveolar lateral flap
ts	[t vJ s]	Affricate (tie bar)
ɧ	[xH]	Simultaneous f and x

DIACRITICS

☞ Voiceless [t̚k]	☞ Breathy voiced [t̚u]	☞ Dental [t̪]
☞ Voiced [t̚v]	☞ Creaky voiced [t̚n]	☞ Apical [d̪]
☞^h Aspirated [t̚ h]	☞ Linguolabial [ɫ]	☞^h Laminar [t̚ d̪]
☞ More rounded [t̚w]	☞^w Labialized [t̚ w]	☞ Nasalized [t̚ ñ]
☞ Less rounded [t̚w̥]	☞^j Palatalized [t̚ j]	☞ⁿ Nasal release [t̚ z̪ n̪]
☞ Advanced [t̚ =]	☞^y Velarized [t̚ g̟ ɤH]	☞^l Lateral release [t̚ z̪ l̪]
☞ Retracted [t̚ =]	☞^ɤ Pharyngealized [t̚ ɜ̞ ɤH]	☞^ɤ No audible release [t̚ p̠]
☞ Centralized [t̚ u]	☞ Velarized or pharyngealized [t̚ ɤy]	
☞^x Mid-centralized [t̚ x]	☞ Raised [t̚ ɤg]	
☞ Syllabic [t̚ s̪]	☞ Lowered [t̚ g̟]	
☞ Non-syllabic [t̚ o]	☞ Advanced tongue root [t̚ h̟]	
☞ Rhoticity [t̚ r̠]	☞ Retracted tongue root [t̚ h̠]	

VOWELS



Where symbols appear in pairs, the one to the right represents a rounded vowel.

SUPRASEGMENTALS

- ◌' ◌^ˈ Primary stress
- ◌◌' ◌◌^ˈ Secondary stress
- ◌◌◌' ◌◌◌^ˈ Long
- ◌◌◌◌' ◌◌◌◌^ˈ Half-long
- ◌◌◌◌◌' ◌◌◌◌◌^ˈ Extra-short
- ◌◌◌◌◌◌ ◌◌◌◌◌◌ Minor (foot) group
- ◌◌◌◌◌◌◌ ◌◌◌◌◌◌◌ Major (intonation) group
- ◌◌◌◌◌◌◌◌ ◌◌◌◌◌◌◌◌ Syllable break
- ◌◌◌◌◌◌◌◌◌◌ ◌◌◌◌◌◌◌◌◌◌ Linking (absence of a break)

TONES AND WORD ACCENTS

LEVEL	CONTOUR
$\hat{\circ}$ v+e Extra high $\circ\uparrow$ v5 Extra high (tone letter)	$\check{\circ}$ v\ Rising
$\acute{\circ}$ ve High $\circ\uparrow$ v4 High (tone letter)	$\hat{\circ}$ v\ Falling
$\tilde{\circ}$ va Mid $\circ\uparrow$ v3 Mid (tone letter)	$\check{\circ}$ ve ve e High rising
$\hat{\circ}$ v' Low $\circ\downarrow$ v2 Low (tone letter)	$\check{\circ}$ v'e' v' Low rising
$\check{\circ}$ v'e' Extra low $\circ\downarrow$ v1 Extra low (tone letter)	$\check{\circ}$ va ve va Rising-falling
$\circ\downarrow$ v6 Downstep $\circ\uparrow$ v+6 Upstep	$\circ\nearrow$ v+7 Global rise $\circ\searrow$ v7 Global fall